the duel, the maths behind what doesnt work

numbers are the films own unless i say its a guess

1. destroyer gun, energy

m is what moves, v is muzzle speed. rod is 19.4 kg, sabot and armature are a guess at about 10 kg

$$KE_{\text{rod}} = \tfrac{1}{2}mv^2 = 0.5 \times 19.4 \times 7800^2 = 590\,\text{MJ}$$
$$KE_{\text{all}} = 0.5 \times 30 \times 7800^2 = 913\,\text{MJ}$$

railguns get about 30 to 40% of the stored energy into the round (η)

$$E_{\text{stored}} = KE/\eta = 913/0.4 \text{ to } 913/0.3 = 2280 \text{ to } 3040\,\text{MJ}$$
$$\text{film bank} = 80 \times 2\,\text{MJ} = 160\,\text{MJ}$$
$$2280/160 = 14\times \text{ short}, \qquad 3040/160 = 19\times \text{ short}$$

even at 100% its $913/160 = 5.7\times$ short. the real navy gun was about 32 MJ

2. acceleration

L is bore travel, 11.61 m

$$a = \frac{v^2}{2L} = \frac{7800^2}{2 \times 11.61} = 2.62 \times 10^6\,\text{m/s}^2 = 267{,}000\,g$$
$$t = \frac{2L}{v} = \frac{23.22}{7800} = 3\,\text{ms}$$

a tank round is about 30,000 g so this is about 9 times that

3. rod gets crushed

the sabot only holds the middle, the nose sticks out 0.52 m and the tail 0.38 m. ρ is tungsten, 19300 kg/m^3

$$\sigma = \rho L a$$
$$\sigma_{\text{nose}} = 19300 \times 0.52 \times 2.62 \times 10^6 = 26.3\,\text{GPa (squashed)}$$
$$\sigma_{\text{tail}} = 19300 \times 0.38 \times 2.62 \times 10^6 = 19.2\,\text{GPa (pulled)}$$

tungsten alloy gives at 1 to 1.5 GPa so thats 18 to 26 times over

the lattice sabot has the armature pushing its back face with the rod and itself on it, about 27 kg

$$p = \frac{F}{A} = \frac{27 \times 2.62 \times 10^6}{\pi \times 0.14^2} = \frac{70.7 \times 10^6}{0.0616} = 1.15\,\text{GPa}$$

a big metal lattice holds maybe 50 to 100 MPa, so 11 to 23 times over

4. field and current

bore is 0.28 by 0.28 m, $\mu_0 = 1.2566 \times 10^{-6}$

$$p = \frac{F}{A} = \frac{30 \times 2.62 \times 10^6}{0.0784} = 1.0\,\text{GPa}$$
$$B = \sqrt{2\mu_0 p} = \sqrt{2 \times 1.2566 \times 10^{-6} \times 10^9} = 50\,\text{T}$$

strongest steady magnet ever made is about 45 T. L' is rail inductance per metre, about $0.45\,\mu$H/m, w is bore width, 10 T from the coils

$$\text{rails only: } I = \sqrt{2F/L'} = \sqrt{\frac{2 \times 78.6 \times 10^6}{0.45 \times 10^{-6}}} = 18.7\,\text{MA}$$
$$\text{with coils: } F = \tfrac{1}{2}L'I^2 + IBw \;\Rightarrow\; 78.6 \times 10^6 = 0.225 \times 10^{-6} I^2 + 2.8I$$
$$I = \frac{-2.8 + \sqrt{2.8^2 + 4 \times 0.225 \times 10^{-6} \times 78.6 \times 10^6}}{2 \times 0.225 \times 10^{-6}} = 13.5\,\text{MA}$$

so the coils do help, the problem is what the rails do to them. coil at $x = 0.51$, rails at $y = \pm 0.19$

$$d = \sqrt{0.51^2 + 0.19^2} = 0.544\,\text{m}$$
$$B_{\text{coil}} = 2 \times \frac{2 \times 10^{-7}\, I}{d} \times \frac{0.19}{0.544} = 4.96 \times 0.698 = 3.5\,\text{T}$$
$$\frac{dB}{dt} = \frac{3.5}{0.003} = 1150\,\text{T/s}$$

fastest superconducting magnets ramp about 4 T/s, this is about 300 times that so they quench every shot

5. rails melt

two copper rails, 0.26 wide, 0.10 thick, 12.7 m long. c for copper is 385 J/kg K. lost energy is about $2280 - 913 = 1370$ MJ, say half goes in the rails

$$m = 0.26 \times 0.10 \times 12.7 \times 2 \times 8960 = 5917\,\text{kg}$$
$$\Delta T = \frac{Q}{mc} = \frac{680 \times 10^6}{5917 \times 385} = 298\,\text{K}$$

fine on average, but a 3 ms pulse only runs in a thin skin. ρ_e is copper resistivity 1.7×10^{-8}

$$\delta = \sqrt{\rho_e t/\mu_0} = \sqrt{\frac{1.7 \times 10^{-8} \times 0.003}{1.2566 \times 10^{-6}}} = 6.4\,\text{mm}$$
$$m_{\text{skin}} = 0.26 \times 0.0064 \times 12.7 \times 2 \times 8960 = 379\,\text{kg}$$
$$E_{\text{melt}} = 379 \times \big(385 \times 1058 + 205{,}000\big) = 232\,\text{MJ}$$

$232/680 = 0.34$ so a third of the heat melts the whole inner face. solid armatures also start arcing around 2.5 km/s and $7800/2500 = 3.1\times$ past that

6. superconducting inductors in the modules

superconducting only saves resistive loss. copper inductor resistance is a guess, 0.1 mΩ

$$I_{\text{module}} = 18.7 \times 10^6/80 = 234\,\text{kA}$$
$$\text{loss} = I^2Rt = 234{,}000^2 \times 0.0001 \times 0.003 = 16.4\,\text{kJ}$$
$$16.4\,\text{kJ}/2\,\text{MJ} = 0.8\%$$

saves under 1% and quenches like the coils

7. orbital ship capacitors

pulse capacitors hold about 1 to 2 kJ/kg

$$m = \frac{6 \times 10^{11}}{2000} \text{ to } \frac{6 \times 10^{11}}{1000} = 300{,}000 \text{ to } 600{,}000\,\text{t}$$

thats 3 to 6 aircraft carriers. a 20 T magnet store would be

$$u = \frac{B^2}{2\mu_0} = \frac{400}{2.513 \times 10^{-6}} = 159\,\text{MJ/m}^3, \qquad V = \frac{6 \times 10^{11}}{1.59 \times 10^8} = 3770\,\text{m}^3$$

8. orbital aiming

the rod keeps the ships orbital speed. $\mu = 3.986 \times 10^{14}$, $R = 6371$ km, $r = 6751$ km

$$v_{\text{orbit}} = \sqrt{\mu/r} = \sqrt{3.986 \times 10^{14}/6.751 \times 10^6} = 7684\,\text{m/s}$$

the gun adds 5000 so even fired straight back its still doing 2684 forward

$$v_{\text{imp}}^2 = v_0^2 + 2\mu\left(\tfrac{1}{R} - \tfrac{1}{r}\right), \qquad 2\mu\left(\tfrac{1}{R} - \tfrac{1}{r}\right) = 7.04 \times 10^6$$
$$h = r\,v_{\text{horiz}}, \qquad v_h = h/R, \qquad \cos\gamma = v_h/v_{\text{imp}}$$

h is angular momentum per kg (stays the same), γ is the angle down from the horizon at impact

straight down: $v_0^2 = 7684^2 + 5000^2 = 8.40 \times 10^7$ $v_{\text{imp}} = 9544$, $\cos\gamma = 8142/9544$, $\gamma = 31°$

30° below back: $v_0^2 = 3354^2 + 2500^2 = 1.75 \times 10^7$ $v_{\text{imp}} = 4954$, $\cos\gamma = 3554/4954$, $\gamma = 44°$

straight back: $v_0^2 = 2684^2 = 7.20 \times 10^6$ $v_{\text{imp}} = 3775$, $\cos\gamma = 2844/3775$, $\gamma = 41°$

it cant come in steeper than about 45°, the film has it coming down steep in a straight line. the films formula leaves out the 7684

$$v_{\text{film}} = \sqrt{5000^2 + 2 \times 9.81 \times 380{,}000} \times 0.86 = 5697 \times 0.86 = 4900\,\text{m/s}$$

close to 4954 by luck. a sim of the real fall lands it 400 to 800 km ahead of the point under the ship so it has to fire early and point mostly backwards, not at the destroyer